Mestrado (Integrado) em Engenharia Informática
Universidade Nova de Lisboa

Declarative Problem Solving
Project (part 2) – Packing Squares into Rectangles

Group 1

Tackling the same problem as in the first part, provide first a direct encoding for the problem of packing squares into rectangles (you may recall the details from the first part available on CLIP).

Group 2

Now, solve the same problem again taking advantage of the techniques taught in the classes using e.g. an imperative language to control the process (optionally with multi-shot solving).

Group 3

Briefly compare the two solutions in between each other in terms of performance. Also, compare with your previous solution in the first part of the project.

Deliverables

You should send a single zip file labelled with your student numbers (in groups of max. two students) to `mkn@fct.unl.pt` until **June 10** (in Portugal). The file should contain:

- One folder for each solution developed, labelled appropriately.
- Any test files you may have used, labelled appropriately.
- Any relevant additional material you may have created.
- A short report (single PDF file) which should contain what you did (with references to the corresponding files), commenting also on encoding optimizations applied (if any), and the brief comparison from Group 3, as well as any supplementary material.

The usage of AI-based tools (LLMs etc.) is permitted (e.g. for aiding to write the report) as long as it is not excessive and the usage is concisely documented in the report.

Collaboration between different groups is not permitted. In case of plagiarism all groups involved will fail the course.